

TURNU-MAGURELE, Romania

WMO: 154900

Lat: 43.7603N

Lon: 24.8783E

Elev: 32

StdP: 100.95

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-9.8	-14.8	1.0	-12.4	-11.8	1.4	-9.2	10.0	1.0	8.1	1.2	0.8	320	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	34.7	22.6	33.1	21.9	31.7	21.2	23.9	32.6	23.0	30.9	22.2	29.5	1.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.4	16.1	26.5	20.6	15.3	26.0	19.9	14.7	25.4	71.6	32.7	68.2	30.9	65.4	29.5	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.0	4.7	-16.4	37.9	3.4	1.9	-18.9	39.2	-20.9	40.3	-22.9	41.3	-25.4	42.7		
(5)			-16.6	25.8	3.4	1.3	-19.1	26.7	-21.0	27.5	-22.9	28.3	-25.4	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-0.8	3.4	7.7	13.0	18.1	22.4	24.2	24.1	19.5	12.6	6.3	1.1		
	DBStd	9.54	4.84	5.01	4.12	3.88	3.28	2.83	2.55	2.90	3.93	4.17	4.44	4.71		
	HDD10.0	1057	334	187	94	17	0	0	0	0	22	128	275			
	HDD18.3	2664	592	418	331	164	44	3	0	2	35	181	361	533		
	CDD10.0	2037	0	3	21	107	251	373	441	437	284	103	17	1		
	CDD18.3	604	0	0	0	3	37	127	183	181	69	3	0	0		
	CDH23.3	6253	0	0	6	75	404	1261	1859	1894	690	63	2	0		
	CDH26.7	2629	0	0	1	14	96	507	842	896	263	11	0	0		
(14) Wind	WSAvg	1.6	1.5	1.8	1.9	1.9	1.6	1.4	1.4	1.3	1.4	1.4	1.5	1.7		
(15) Precipitation	PrecAvg	515	34	27	33	46	57	61	55	34	47	45	32	39		
	PrecMax	868	110	66	80	129	145	133	172	113	138	91	81	154		
	PrecMin	284	4	2	1	14	17	17	13	4	0	0	0	2		
	PrecStd	119	22	17	21	24	27	27	40	24	33	28	23	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	19.1	23.2	28.5	31.1	35.8	36.2	36.6	34.5	28.3	22.1	14.4		
		MCWB	7.5	12.9	13.7	17.8	20.4	23.8	23.7	23.2	22.2	18.8	15.1	9.9		
	2%	DB	9.3	15.8	20.2	25.4	29.0	33.2	34.3	34.7	31.8	24.9	17.9	11.8		
		MCWB	6.3	10.6	12.3	16.4	19.3	22.3	22.9	22.1	19.9	16.8	13.5	8.7		
	5%	DB	7.0	13.0	17.8	22.7	27.4	31.3	32.7	33.1	29.6	22.4	15.0	9.1		
		MCWB	5.0	9.3	10.9	14.9	18.4	21.3	22.3	21.4	19.2	15.8	11.6	6.8		
	10%	DB	4.9	10.5	15.2	20.6	25.5	29.7	31.2	31.5	27.6	19.9	12.5	6.9		
		MCWB	3.9	7.8	9.7	13.7	17.6	20.7	21.6	21.0	18.8	14.7	10.5	5.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	13.3	14.4	18.2	21.5	24.5	25.0	24.6	23.7	19.7	15.4	10.8		
		MCDB	11.1	18.8	22.4	26.7	29.8	33.3	33.9	33.8	33.7	25.1	20.8	13.1		
	2%	WB	6.5	11.1	12.6	17.0	20.3	23.0	24.0	23.5	21.6	17.5	13.7	8.9		
		MCDB	8.7	14.8	19.0	24.4	27.6	31.2	32.2	32.2	28.9	23.1	17.9	11.8		
	5%	WB	5.3	9.5	11.4	15.7	19.0	22.1	23.2	22.5	20.3	16.3	12.3	7.2		
		MCDB	6.7	12.6	17.1	21.8	25.7	29.7	30.8	30.4	27.3	21.6	14.8	9.0		
	10%	WB	3.9	8.1	10.1	14.4	18.0	21.4	22.5	21.7	19.2	15.1	10.9	5.6		
		MCDB	4.9	10.3	14.5	19.3	24.2	28.3	29.4	29.2	25.5	19.3	12.6	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	9.0	10.5	11.7	12.0	12.3	12.3	13.0	12.5	11.2	7.2	6.6		
		MCDBR	10.1	14.2	16.6	16.5	14.9	15.0	14.9	15.3	16.2	15.8	12.5	10.7		
		MCWBR	7.1	9.1	9.2	8.1	6.3	5.6	5.6	5.3	6.4	8.1	7.6	7.4		
	5% WB	MCDBR	8.9	13.0	15.3	15.5	13.7	14.0	13.3	13.6	14.5	14.6	11.1	10.1		
		MCWBR	6.5	8.6	8.5	7.7	6.1	5.5	5.2	4.9	6.3	7.6	6.8	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.347	0.377	0.415	0.471	0.447	0.453	0.464	0.478	0.443	0.404	0.373	0.351		
	taud		2.285	2.212	2.160	2.018	2.165	2.184	2.160	2.117	2.180	2.275	2.317	2.301		
	Ebn at Noon		756	795	813	795	830	824	808	780	776	757	720	710		
	Edh at Noon		87	110	132	165	146	144	146	148	128	103	84	78		
(44) All-Sky Solar Radiation	RadAvg		1.46	2.29	3.46	4.75	5.86	6.45	6.52	5.81	4.22	2.74	1.55	1.20		
	RadStd		0.22	0.31	0.34	0.48	0.47	0.42	0.41	0.37	0.33	0.33	0.22	0.17		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.47	N/A	N/A	N/A	+0.40	N/A	-143	+65	N/A

Nomenclature: See separate page